EE4402 Tutorial One

Q1. Let $\hat{x}$(t) is the Hilbert transform of a real signal $x(t)$. Prove the results follows.

a) If $x(t)$ is even symmetric, $\hat{x}$(t) is odd symmetric. If $x(t)$ is odd symmetric, $\hat{x}$(t) is even symmetric.

b) If $x(t) = cos 2\pi f_0 t$, then $\hat{x}(t) = sin 2\pi f_0 t$.

c) $\hat{\hat{x}}(t) = -x(t)$.

d) $\int_{-\infty}^{\infty} x^2(t)dt = \int_{-\infty}^{\infty} \hat{x}^2(t)dt$.

e) $x(t)$ and $\hat{x}(t)$ are orthogonal.

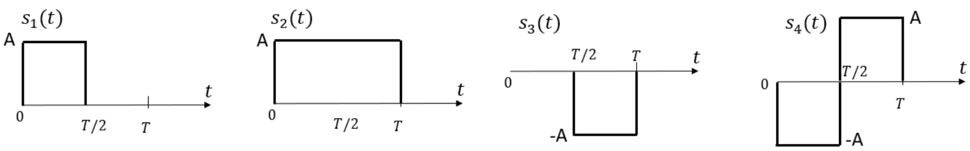
Q2. Assume T is a positive constant and $x_i \in \{0,1\}$ for $0 \le i \le 2$. [Hint: $\mathrm{sinc}\left(\frac{t}{T}\right) \leftrightarrow T \times \Pi(fT)$]

a) Find energy of the signal $\mathrm{sinc}(t)$.

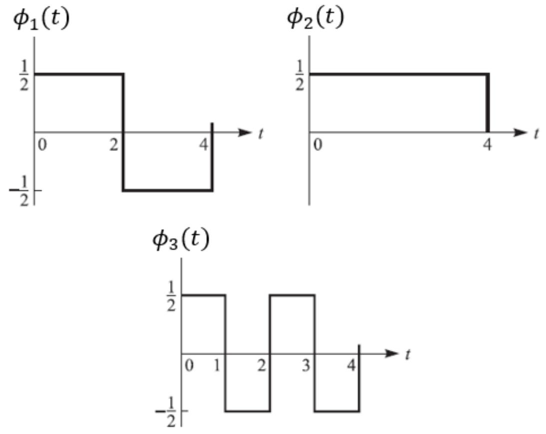
b) Show that $\mathrm{sinc}\left(\frac{t}{T}\right)$ and $\mathrm{sinc}\left(\frac{t-T}{T}\right)$ are orthogonal.

c) Propose a set of orthonormal basis functions for signals $s(t) = \sum_{i=0}^{2} x_i \times \mathrm{sinc}\left(\frac{t-iT}{T}\right)$. Find the size of the signal set $\{s(t)\}$ and its average energy assuming equiprobable signals.

Q3. Find a set of orthonormal basis functions for the following signal set. Find their vector forms and sketch their constellation.



Q4. The waveforms for three functions $\phi_i(t)$, $1 \le i \le 3$, are given as below.



a) Show they are orthonormal.

b) The signal

$$x(t) = \begin{cases} -1, & 0 \le t < 1 \\ 0, & 1 \le t < 3 \\ 1, & 3 \le t < 4 \end{cases}$$

1) Find the energy $\mathcal{E}_x$ of signal $x(t)$.

2) Project $x(t)$ to the signal space constructed by $\phi_i(t)$, $1 \le i \le 3$. Determine the vector form $\boldsymbol{x}$ of $x(t)$.

3) Find the energy of $x(t)$ using $\boldsymbol{x}$.

Q5. The Q function is defined as $Q(x) = \int_x^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt$.

a) Show that $Q(x) \le \frac{1}{2} e^{-\frac{x^2}{2}}$ when $x \ge 0$.

b) For all $x \ge 0$, show that $\frac{x}{\sqrt{2\pi}(1+x^2)} e^{-\frac{x^2}{2}} < Q(x) < \frac{1}{x\sqrt{2\pi}} e^{-\frac{x^2}{2}}$.

Q6. Let random variable $X \sim \mathcal{N}(0, \sigma^2)$. If another random variable is generated by $Y = g(X)$. Find the PDF or PMF of Y for following cases.

a) $g(x) = \begin{cases} -b, & x \leq -b \\ b, & x \geq b \\ x, & |x| < b \end{cases}$

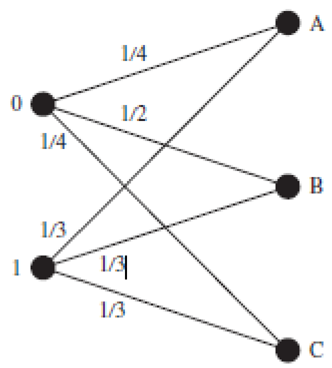
b) $g(x) = \begin{cases} a, & x > 0 \\ 0, & x = 0 \\ b, & x < 0 \end{cases}$

EE4402 Tutorial Two

Main topics in Chapter 3 and Chapter 4

1. Linear modulations.
2. High dimensional modulations.
3. Optimal receiver in AWGN channels.
4. Optimal receiver performance in AWGN channels.
5. Implementation of optimal receiver for AWGN channels.

Q1. A communication channel with binary input and ternary output alphabets is shown in Figure below. The probability of the input being 0 is 0.4. The transition probabilities due to the channel distortion are shown on the figure.

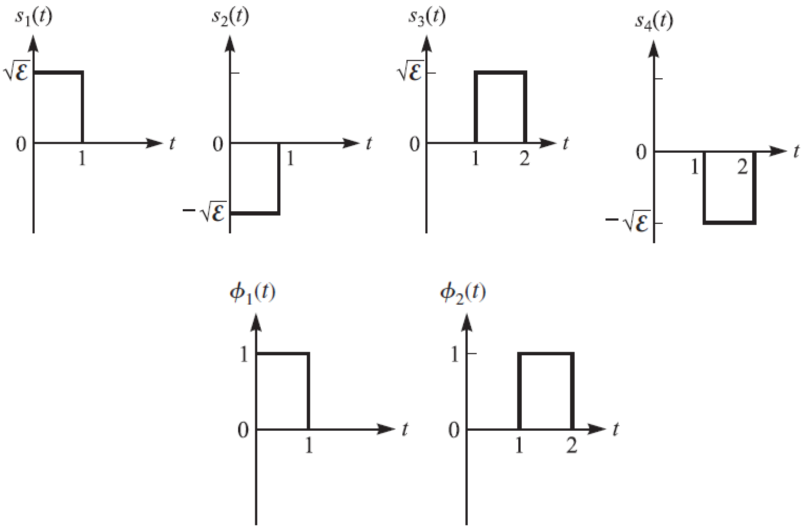


a) For each of the channel outputs A, B and C, what is the best decision on channel input that minimizes the error probability?

b) If a 0 is transmitted and the optimal decision scheme derived in a) is applied at the receiver, what is the probability of error?

c) What is the overall error probability for this channel if the optimal decision scheme is applied at the receiver?

Q2. A modulation scheme uses the four signals $s_k(t)$, $k = 1, 2, 3, 4$, shown in the figure below, with the orthonormal functions $\phi_1(t)$ and $\phi_2(t)$ as the basis functions.



a) Determine the vector forms of $s_k(t)$ for $k = 1, 2, 3, 4$.

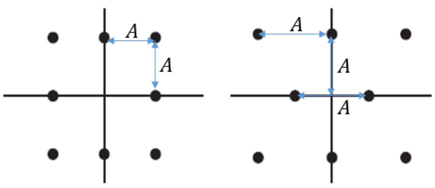
b) Plot the signal constellation and label the signal points using Gray coding.

c) What is this modulation based on the signal set?

d) Find the minimum Euclidean distance, average signal energy and average bit energy?

e) If the signal set is transmitted in an AWGN channel and all signals are equiproble, what is the optimal detection rule?

Q3. Consider the two 8-point QAM signal constellations shown in figure below. The minimum distance between adjacent points is A which is a positive constant.



(a) (b)

a) Find the vector form of each signals in two constellations.

b) If all signals are equiprobable, determine the average signal energy for each constellation and indicate which constellation is more power-efficient?

c) If the communication is in an AWGN channel with noise variance $\frac{N_0}{2}$, estimate the symbol error rate using the union bound and simplify the union bound in terms of minimum Euclidean distance.

Q4. A binary digital communication system employs the signals

$$s_0(t) = 0, \;\; 0 \leq t \leq T_S$$

$$s_1(t) = A, \;\; 0 \leq t \leq T_s$$

for transmitting the information. This is called on-off signaling. The demodulator cross correlates the received signal $r\,(t)$ with $s_0(t)$ and $s_1(t)$, respectively, and samples the output of the correlator at $t = T_S$.

a) If all signals are equiprobable, determine the optimum detector for an AWGN channel.

b) Determine the probability of symbol error as a function of $\frac{E_b}{N_0}$ if both signals are transmitted with equal probability.

c) Compare performance of the on-off signaling and the antipodal signaling. (Assume both are with equiprobable signals and compare their SER vs $\frac{E_S}{N_0}$.)

Q5. A ternary communication system transmits one of three real equiprobable signals $s(t), 0$, or $-s(t)$ every T seconds. The received signal is $r(t) = s(t) + z(t), r(t) = z(t)$, or $r(t) = -s(t) + z(t)$, where $z(t)$ is white Gaussian noise with sample variance $N_0/2$. The energy of $s(t)$ is E_s. The optimum receiver computes the correlation metric

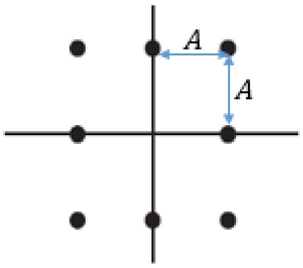
$$U = \int_0^T r(t)s(t)dt$$

and compares U with a threshold $A = \frac{E_S}{2}$ and a threshold $-A$. If $U > A$, the decision is made that $s(t)$ was sent. If $U < -A$, the decision is made in favor of $-s(t)$. If $-A < U < A$, the decision is made in favor of 0. Note that $N = \int_0^T n(t)s(t)dt$ is a zero mean Gaussian random variable of variance $\frac{E_S N_0}{2}$.

a) Determine the three conditional probabilities of error: P_e given that $s(t)$ was sent, P_e given that $-s(t)$ was sent, and P_e given that 0 was sent.

b) Determine the average probability of error P_e as a function of the threshold A, assuming that these three symbols are equiprobable.

Q6. The signal constellation for a communication system with 8 equiprobable symbols is shown in figure below. The channel is AWGN with noise variance $N_0/2$.



a) Using the union bound, find a bound in terms of A and N_0 on the error probability for this channel.

b) Determine the average SNR per bit for this channel.

c) Express the bound found in a) in terms of the average SNR per bit.

Q7. Two signals $s_1(t) = 3d\cos(2\pi f_0 t)$, $s_2(t) = 4d\sin(2\pi f_0 t)$, where $0 \le t \le T_S$, are used as the signal set for digital transmission in an AWGN channel with noise variance $\frac{N_0}{2}$. T_s is the symbol duration and assume $T_s \times f_0$ is an integer.

a) Indicate the basis function(s) of the signal set and their vector representations under this set of basis function(s).

b) Find the average bit energy and the signal-to-noise ratio $\gamma_b = E_b/N_0$.

c) Derive the decision boundary for the receiver to minimize the bit error probability and show it in a sketch of the constellation.

d) Calculate the bit error rate and express it using Q function and γ_b.

Q8. The discrete sequence

$$r_k = \sqrt{\mathcal{E}} c_k + n_k \ , k = 1, 2, \ldots, n$$

represents the output sequence of samples from a demodulator, where $c_k = \pm 1$ are elements of one of two possible codewords, $c_1 = [1\ 1\ \cdots\ 1]$ and $c_2 = [1\ 1\ \cdots\ 1\ -1\ \cdots\ -1]$. The code word c_2 has w elements that are $+1$ and $n - w$ elements that are -1, where w is some positive integer or zero. The noise sequence $\{n_k\}$ is white Gaussian with variance $\frac{N_0}{2}$. Assume two code words are equiprobable.

a) What is the optimum maximum-likelihood detector for the two possible transmitted signals?

b) Determine the probability of error as a function of the parameters $(\frac{N_0}{2}, \mathcal{E}, w)$.

c) What is the value of w that minimizes the error?

EE4402 Tutorial Three

Main topics in Chapters 5- 7

1. Signaling bandlimited channel.
2. Linear block codes.
3. Convolutional codes.

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1. An 8-kHz bandpass channel is to be used for transmission of data at a rate of 9600 bits/s. If the additive zero-mean Gaussian noise sample variance is $\frac{1}{2}N_0 = 10^{-10}$ in the channel, design following modulations and determine the average power that achieves a bit error probability of 10^{-6} using a signal pulse with a raised cosine spectrum having a roll-off factor of at least 50 percent.

   a) PAM.

   b) PSK.

2. Let C be a binary $(6, 3)$ code with parity check matrix,

$$H = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

   a) What is the minimum Hamming distance of this code?

   b) Find the generator matrix G.

   c) Encode message $\boldsymbol{m} = (1\ 01\ )$.

   d) Suppose the code is used for error detection over a binary symmetric channel with crossover probability $p = 10^{-3}$. Find the probability of undetected error.

   e) Decode the noisy code word $r = (\times \times \times\ 1\ 1\ 0)$ where $\times$ means an unknown value.

3. Let C be the binary repetition code of length 5.

   a) List the vectors which will be decoded as 11111 using Hamming distance rule.

   b) If this code is used in a BSC channel and the crossover probability is p, find the proability P_e of a noisy code word is decoded incorrectly?

   c) When $p = 0.05$, what is the P_e?

4. Prove that it is not possible to find 32 binary words, each of length 8 bits, such that each code word differs from every other code word in at least 3 places.

5. A binary (2,1,2) convolutional code with generator $g_0 = (101)$ and $g_1 = (1\ 1\ 1)$.

   a) Write down the parity check equations.

   b) Sketch the encoder block diagram and encode the message (1 0 1 1 1).

   c) Find its state machine and trellis diagram. Encode the message (1 0 1 1 1).

   d) Find the free distance of this code.

   e) Use Viterbi decoder to find the decoding output of received sequence (11 11 01 10 01 10 11).